Jiangyou Zhu

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EDUCATION

North University of China **Engineering Bachelor** **Sep.2017-Jun.2021**

- **GPA: 4.2/5.0, ranking 1/163**. Average Score: **90.62/100**
- Honor: National Scholarship for Undergraduate Excellence (2019)
  First-class Scholarship for Excellent Students (2018-2021)

Beihang University **Engineering Master** **Sep.2021-Jun.2024**

- **GPA: 3.8/4.0, ranking: 2/145**. Average Score: **91.86/100**
- Honor: National Scholarship for Graduate Excellence (2023)
  First-class Scholarship for Excellent Students (2022-2024)

National University of Singapore

- School of Computing **Artificial Intelligence and Machine Learning Course** **Jul.2023-Aug.2023**

SCIENTIFIC RESEARCH

ACADEMIC PAPER

1. **Jiangyou Zhu**, Peng Lei, et al., "**Grating Lobe Suppression and Angle Estimation Based on Virtual Antennas Filling in Sparse Array**," ***IEEE Antennas and Wireless Propagation Letters***., vol. 22, no. 3, pp. 502–506, **Mar. 2023**.

- Novelty: Aiming at angle ambiguity problem affected by grating lobes, the proposed method does not require the design of a special array and is suitable for sparse array where the spacing between two adjacent antennas is uniform or non-uniform. Both simulations and experiments are carried out to verify its effectiveness in different types of sparse arrays.
- Responsibility: I proposed a grating lobe suppression and angle estimation approach for sparse arrays by use of virtual antennas filling. It takes advantage of the distribution characteristics of grating lobes, and the gaps between antenna elements of sparse array could be filled by using least squares method. Hence, it achieves the grating lobe suppression as well as accurate estimation of target angles.

2. **Jiangyou Zhu**, Peng Lei, Jun Wang, Yaotian Zhang, "**Design and Implementation of Bird Detection Algorithm Based on 24GHz Millimeter-Wave Radar**," in ***Proceedings of the 14th National Conference on DSP Application Technolog (DSP)***, Beijing, China, **Oct. 2022**.

- Novelty: The proposed radar bird detection system meets the requirements of power system miniaturization, short distance and low cost in real life, which can satisfy the measurement accuracy and realize large-scale application.
- Responsibility: I proposed a design idea of a bird detection algorithm based on 24GHz millimeter-wave radar; at the same time, according to its motion characteristics, and proposed a multi-dimensional information fusion method for identifying flying birds and eliminating false targets, which effectively improves the ability of the algorithm to remove clutter. At present, this algorithm has been applied to actual power pole and tower systems.

3. Chi Zhang, Mingming Jin, **Jiangyou Zhu**, Ge Dong, and Bin Yang, "**TDM Radar Target Angle and Velocity Estimation with Forward and Reverse Transmission Patterns**," in ***IET International Radar Conference 2023***, Chongqing, China, Dec. 2023. **(Accepted)**

- Novelty: To address velocity and angle coupling in TDM-MIMO radar systems for moving targets, a forward and reverse transmission pattern is proposed together with the parameter estimation method is proposed.
- Responsibility: I proposed forward and reverse transmission modes for radar operation. This allows extracting target velocity and angle phases, enabling solving independent parameters from the coupled phase.

4. Yuanhao Wang, Guidong He, Tieshuai Song, **Jiangyou Zhu**, Chi Zhang, Xinyuan He and Jun Wang, "**Enhancing 3D SAR Imaging: a Near-field Back Projection Algorithm for Addressing Geometric Distortion**," in ***2024 IEEE International Geoscience and Remote Sensing Symposium***, Athens, Greece, Jul. 2024. **(Accepted)**

- Novelty: This paper introduces a novel MIMO SAR algorithm to correct geometric distortion in 2D SAR under near-field conditions, enhancing performance by utilizing the true locations of transmit and receive arrays.
- Responsibility: My responsibilities encompass the division of the imaging area into grids and the implementation of phase compensation. This entails partitioning the area into grids of optimal size, computing the distances between each grid point and the radar, and subsequently applying phase compensation techniques to achieve accurate back-projection.

NATIONAL INVENTION PATENT

1. Yaotian Zhang, **Jiangyou Zhu**, et al., **Aperture expansion method of motion synthesis array based on generalized co-prime array**, accepted, application number: 2023104502059, **Apr. 2023**. **(Student First Inventor)**

- Novelty: This invention introduced a motion synthetic co-prime array to improve the DOF and the angle accuracy.
- Responsibility: I proposed the virtual array interpolation technology to make full use of the information of target signal and proposed an off-grid model to reduce the anglet error.

2. Yaotian Zhang, **Jiangyou Zhu**, et al., **Motion-assisted sparse linear array millimeter-wave radar angle estimation method**, accepted, application number: 2023111134460, **Aug. 2023**. **(Student First Inventor)**

- Novelty: This invention does not need to design a sparse array, can save computing power and reduce mutual coupling.
- Responsibility: I proposed a method of angle estimation for millimeter-wave radar with sparse uniform array assisted by motion information of target to solve the angle ambiguity problem.

PROJECT EXPERIENCE

1. National Natural Science Foundation of China **Jan.2023-May.2023**

- ✧ Title: Angle Super-resolution Method for MIMO Radar Virtual Launch and Waveform Optimal Scheduling
- ✧ Responsibility: I completed the project basis and research content, and proposed a virtual transmit aperture expansion method based on joint optimization of amplitude, phase, and transmit array elements.

2. Research and development project of Zhejiang Province **Apr.2022-Dec.2022**

- ✧ Title: Research on Multi-target Angle Estimation Method for Millimeter-Wave MIMO Radar
- ✧ Responsibility: I proposed a multi-target angle estimation method to solve the problem of inaccurate angle measurement of weak targets caused by adjacent targets in multi-target scenarios.

3. **Leading Innovation and Entrepreneurship Team Construction Project** **Sep.2021-Mar.2022**

- ✧ Title: Gesture recognition system of millimeter wave radar based on deep learning
- ✧ Responsibility: I proposed a gesture recognition method based on the 3D convolutional neural network input of Range-Doppler map and Range-Angle map sequence.

ACADEMIC APPOINTMENT

- ✧ Reviewer of EURASIP Journal on Advances in Signal Processing SCI journal, IEEE student member.

STUDENT WORK

Master

- ✧ Monitor of Class ZY21025, School of Electronic Information Engineering, Beihang University. **Sep.2021-Jun.2024**

Undergraduate

- ✧ Assistant Teacher, School of Information and Engineering, North University of China. **Sep.2017-Jun.2021**

AWARDS AND HONOR

Master

- ✧ The **National Scholarship**. **Oct. 2023**
- ✧ The **First-class Scholarship of Beihang University**. (**the highest award for three consecutive years**).
- ✧ The **Outstanding Graduate Student** of Beihang University in 2022. Jan. 2023
- ✧ The **Merit graduate student** of Beihang University in 2022. Oct. 2022
- ✧ The National Second Prize of the China Postgraduate Mathematical Contest in Modeling (**ranked 2nd**). Jan. 2023
- ✧ The National Second Prize of IC and Electronic Design Invitational Competition for Cross-Strait College Students (**ranked 1st**). Sep. 2022
- ✧ The Third Prize of the 2022 China Graduate Electronic Design Competition (**ranked 1st**). Aug. 2022
- ✧ The National Second Prize of National Electronic Design Invitational Competition for Cross-Strait College Students (**ranked 2nd**). Dec. 2021

Undergraduate

- ✧ The **Student Star representative** of North University of China in 2021. Jun. 2021
- ✧ The Provincial Third Prize of China College Student Computer Design Competition (ranked 2nd). Jun. 2020
- ✧ The **National Scholarship**. **Dec. 2019**
- ✧ The **First Prize** of the National Mathematical Contest in Modeling for College Students (**ranked 1st**). Dec. 2019
- ✧ The **First prize** of Computer Application Competition of five provinces (municipalities and autonomous regions) in North China and Hong Kong, Macao and Taiwan College students (ranked 5th). Nov. 2019
- ✧ The **Gold Medal** of Shanxi Province Innovation and Entrepreneurship Competition for College Student. Oct. 2019
- ✧ The National College Student Innovation Project. Sep. 2019
- ✧ The **First Prize** of the National College Students Electronic Design Competition (**ranked 1st**). Sep. 2019
- ✧ The **First Prize** of Academic English Vocabulary Competition for College Students (**ranked 1st**). Jul. 2019
- ✧ The **Second Prize** of National Software and Information Technology Professionals Competition. Apr. 2019
- ✧ The **Freshman Scholarship** for undergraduate excellent of North University of China. Sep. 2017
- ✧ The **First-class Undergraduate Scholarship** (**the highest award for four consecutive years**).

PROFESSIONAL SKILL

- ✧ **Foreign language level:** The overall score band of the **IELTS** result is **6.5**, specifically listening 6, reading 7, writing 6 and speaking 6.
- ✧ **Technical ability:** proficient in using MATLAB for algorithm simulation, have passed the National Computer Level Examination Level 2, Level 3, and Level 4.
- ✧ **Writing ability:** write project documents and related test summary documents for many times, with strong document writing ability. can proficiently write English scientific papers.
- ✧ **Cooperation ability:** As an ATR State Key Laboratory member, I've led multiple millimeter-wave radar tests and collaborated on diverse target identification tasks, showcasing robust communication and teamwork abilities.
- ✧ **Scientific research attitude:** love scientific research, serious and responsible, studious, hard-working, strong ability to work under pressure and adaptability.